

Tz-Hsuan TSENG

[Google Scholar](#)

zengzx22@mails.tsinghua.edu.cn

<https://bgc.yale.edu/people/tz-hsuan-tseng-ms>

Education

Visiting Assistant in Research. Yale University (PI: Walter Jetz) Ecology & Evolutionary Biology	Oct 2024 – Oct 2025
Ph.D. Candidate. Tsinghua University (PI: Rui Yang) Landscape Architecture (Beijing, China)	2022 – Present
M.Sc. Tsinghua University Landscape Architecture (Beijing, China)	2019 – 2022
B.Sc. Nanjing University Geography (Nanjing, China)	2015 – 2019

Professional experiences

Gaoligong Mountains National Nature Reserve, Intern	Feb. 2026
IUCN WCPA, Research Officer	2025 – Present

Peer-reviewed publications

1. **Tseng, T.H.**, Wang, F., Sun, M., Cao, Y., Hou, S., Zhou, Y., Zhao, Z. and Yang, R., 2025. Protected areas unevenly contribute to terrestrial vertebrate habitat conservation in China. *Communications Earth & Environment*, 6(1), p.313.
2. **Tseng, T.H.**, Yang, R., Huang, Y., Chen, L., 2025. Characteristics of bird diversity and environmental relationships in Tsinghua University campus. *Biodiversity Science*, 33(5), p.24373. [in Chinese]
3. **Tseng, T.H.**, Wang, F., Yang, R., 2024. The concept of representativeness and research progress in the field of biodiversity conservation. *Journal of Beijing Forestry University Social Sciences*, 23(02):18-27. [in Chinese]
4. **Tseng, T.H.**, Chen, X., Cao, Y., Yang, R., 2021. Progress and suggestions of biodiversity conservation on university campuses. *Environmental Protection*, 49(21), 35-40. [in Chinese]
5. Cao, Y., **Tseng, T.H.**, Wang, F., Jacobson, A., Yu, L., Zhao, J., Carver, S., Locke, H., Zhao, Z. and Yang, R., 2022. Potential wilderness loss could undermine the post-2020 global biodiversity framework. *Biological Conservation*, 275, p.109753.
6. Yang, R., Cao, Y., Hou, S., Peng, Q., Wang, X., Wang, F., **Tseng, T.H.**, Yu, L., Carver, S., Convery, I. and Zhao, Z., 2020. Cost-effective priorities for the expansion of global terrestrial protected areas: Setting post-2020 global and national targets. *Science Advances*, 6(37), p.eabc3436.
7. Hou, S., Yang, R., Zhao, Z., Cao, Y., **Tseng, T.H.**, Wang, F., Wang, H., Wang, P., Wang, X. and Yu, L., 2024. A cost-effective approach to identify conservation priority for 30×30 biodiversity target on the premise of food security. *Science of The Total Environment*, 941, p.172870.
8. Cao, Y., Wang, F., **Tseng, T.H.**, Carver, S., Chen, X., Zhao, J., Yu, L., Li, F., Zhao, Z. and Yang, R., 2022. Identifying ecosystem service value and potential loss of wilderness areas in China to support post-2020 global biodiversity conservation. *Science of The Total Environment*, 846, p.157348.
9. Wang, F., Zhao, Z., Wang, P., Zhong, L., Yang, S., Tang, J., Hou, S., **Tseng, T.H.**, Cao, Y. and Yang, R., 2024. Over 1/4 of China's terrestrial area significantly contributed both to biodiversity conservation and carbon neutrality, requiring protection. *Science of the Total Environment*, 912, p.169070.
10. Zhao, Z., Wang, P., Wang, X., Wang, F., **Tseng, T.H.**, Cao, Y., Hou, S., Peng, J. and Yang, R., 2022. A protected area connectivity evaluation and strategy development framework for post-2020 biodiversity conservation. *Land*, 11(10), p.1670.
11. Peng, Q., Yang, R., Cao, Y., Wang, F., Hou, S., **Tseng, T.H.**, Wang, X., Wang, P., Zhao, Z., Yu, L. and Locke, H., 2021. One-third of lands face high conflict risk between biodiversity conservation and human activities in China. *Journal of Environmental Management*, 299, p.113449.
12. Cao, Y., Hou, S., **Tseng, T.H.**, Wang, X., Wang, F., Zhao, Z., Yang, R., 2020. Biodiversity conservation strategies for the Yellow River basin based on the three conditions framework. *Biodiversity Science*, 28, 1447-1458. [in Chinese]
13. Wang, F., Peng, J., He, X., **Tseng, T.H.**, Yang, R., 2020. An International Comparison of National Park Ranger Systems. *Chinese Landscape Architecture*, 38 (4), 32. [in Chinese]
14. Zhuang, M., Zhang, J., Xiao, X., Qiu, M., Lu, Y., Zhang, H., **Tseng, T.H.**, Zuo, L., Hu, M., 2020. How destination music affects tourists' behaviors: travel with music in Lijiang, China.

Manuscripts in review/preparation

1. **Tseng, T.H.**, Killion, A., Jetz, W., Hidden suboptimality of cost and area minimization in quantitative conservation prioritization model (prepare to submit).

- Hou, S., Yang, R., Ding, Y., **Tseng, T.H.**, Identifying Priority Areas for Biodiversity Inclusive Spatial Planning in Human-dominated Landscapes Based on Birds Extinction Risk Simulation (submitted, *Conservation Biology*).
- P Wang, Z Zhao, R Yang, F, Lin, Y Wang, **Tseng, T.H.**, Extraordinary transboundary dispersal to China by thousands of Mongolian gazelles: response to extreme weather (prepare to submit).
- Cheng, C., **Tseng, T.H.**, Hu, Y., Xie, D., Li, F., Advancing machine learning for scalable ecosystem restoration (prepare to submit).

Presentation/Talk

- Hidden suboptimality of cost and area minimization in conservation prioritization model. International Congress for Conservation Biology (ICCB). Brisbane, Australia. 2025.
- What animals are there in Tsinghua University? Seminar on Nature Conservation Areas on University Campuses. Peking University, China. 2024
- Internal structural features of urban green space affect biodiversity: A study on bird composition in multi-functional campus green space. International Association of Landscape Ecology (IALE) World Congress. Nairobi, Kenya. 2023.
- The contribution of China's protected areas to terrestrial vertebrate habitat conservation. The 2nd Conference of China Conservation Biology. Nanchang, China. 2025.
- The concept of ecological representativeness in protected areas and its research progress. The 1st Conference of China Conservation Biology. Guangzhou, China. 2023.

Funding

Funder – Project name (Role)	Total	Period
Free Flying Wings, SEE Foundation – Bird-Friendly Campus Initiative (Co- Lead).	¥200,000	2025-2026
One Yangtze River Environmental Funding, He Yi Green Foundation (Lead)	¥3,000	2024, 2025
Research Stipend, Yale Center for Biodiversity and Global Change	\$10,125	2025
Scholarship for Overseas Graduate Studies, Tsinghua University	\$10,260	2024-2025

Awards

Best Oral Talk Finalist, ICCB 2025 Student Awards	2025
Special Prize Scholarships for HK, Macao & Taiwan Students. Ministry of Education (¥30,000)	2025
Special Prize Scholarships for HK, Macao & Taiwan Students. Ministry of Education (¥20,000)	2021
First Prize Scholarships for HK, Macao & Taiwan Students. Ministry of Education (¥50,000)	2017, 2018, 2022, 2023
Second Prize Scholarships for HK, Macao & Taiwan Students. Ministry of Education (¥10,000)	2024
Outstanding Master Thesis. School of Architecture, Tsinghua University	2022
Outstanding Undergraduate Thesis. School of Geography and Ocean Science, Nanjing University	2019
Special Prize of Campus Biodiversity Mapping , Friend of Nature, Youth	2021
Second Prize of COP15 Biodiversity Conservation Solution , WWF & Center Environmental Education and Communications of Ministry of Environmental Protection	2021

Applied conservation experiences

Field Research Intern in Gaoligong Mountains National Nature Reserve Feb 2026

Role: Researcher

- Conducted field surveys and semi-structured interviews to investigate the birding economy.
- Documented bird blinds/ponds and birding routes.
- Analyzed citizen science data to support understanding of regional birding tourism.

Tsinghua university campus wildlife-friendly project

2024 – 2026

Client 1: He Yi Green ("One Yangtze River" University Student Environmental Activity Funding), China

Client 2: SEE Foundation

Role: Project Manager; Vice President, Wildlifers Student Association of Tsinghua University

- Designed and implemented a campus wildlife monitoring program using camera traps and acoustic recorders.
- Conducted routine bird surveys, nocturnal bird monitoring, and winter crow counts.
- Led the development of science outreach materials, including a campus biodiversity-themed illustrated picture book.

Survey of Mongolian gazelle (*Procapra gutturosa*) migration in Arxan, Inner Mongolia

2024 – 2025

Role: Assistant, Institute for National Park, Tsinghua University

- Conducted fieldwork along the China–Mongolia border in Arxan.
- Recorded Mongolian gazelle occurrence points; deployed camera traps.

- Performed data analysis to assess the impacts of extreme weather and to identify temporary habitat selection.

Current status and countermeasures for livelihood security in national parks

2023 – 2024

Client: National Forestry and Grassland Administration

Role: Assistant, Institute for National Park, Tsinghua University

- Conducted interviews and questionnaire with local residents and government officials in Giant Panda National Park.
- Analyzed shifts in community perceptions before and after the establishment of the national park system.

Planning major projects for Beijing's protected area system

2023 – 2024

Client: Beijing Municipal Forestry and Parks Bureau, China

Role: Assistant, Institute for National Park, Tsinghua University

- Proposed 5 projects with details of site and action – restoration, corridor, infrastructure, green construction, monitoring
- Co-responsible for smart monitoring planning, identified the needs/gaps, proposed equipment lists.

Consultation on the volunteer system of China's national parks

2023 – 2024

Client: Shan Shui Conservation Center, China

Role: Assistant, Institute for National Park, Tsinghua University

- Contributed to the development of the national workshop and report “Everyone’s National Park: Seminar on Volunteer Participation Mechanisms in National Parks”, including literature review and synthesis of existing research on volunteer systems.
- Participated in the planning of an ecological trail in Diqing Village, Three-River-Source National Park.

Developing national- and local-level guidance on reconciling conservation and development conflicts in China’s protected areas

2022 – 2023

Client: World Bank

Role: Assistant, Institute for National Park, Tsinghua University

- Identified conflict hotspots and its drivers; investigated current status of 5 national parks (NP); proposed actions.
- Responsible for questionnaires and interviews in 3 national parks (Three-River Source, Giant Panda, Hainan Tropical Rainforest), questionnaire analysis, and the writing of field study report.

Beijing urban biodiversity survey

2021

Role: Volunteer, Eco-Bridge Continental

- Conducted mammal surveys across both urban and mountainous areas of Beijing.
- Site selection and deployment of traps and camera traps, and ensured timely retrieval and data recording.

Professional service and outreach

Reviewer for: Nature Communications; Communications Earth & Environment; Ecological Indicators; Landscape Ecology; Ecography; Land Degradation & Development

Teaching assistant: Practice in Landscape architecture; Classic literatures in Landscape Architecture; Tsinghua University Global Open Courses National Parks and Protected Areas of China.

SKILLS

Programming: R, Python, GEE

Software: ArcGIS/QGIS/GeoDa, Marxan, Zonation, Fragstats, SPSS, InVEST, ENVI

Methodology: Conservation prioritization, Spatial analysis, Regression model, Bird/Plant survey, Camera trap setting, Species distribution model, CNN, Content analysis